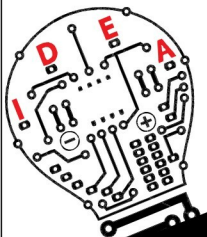


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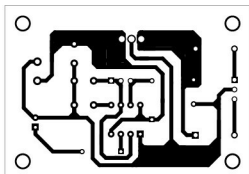


Fig. 3: An actual-size PCB pattern of the transmitter circuit

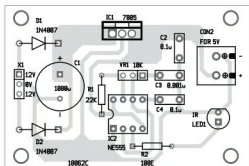


Fig. 4: Component layout for the PCB in Fig. 3

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - NE555 timer
- IC3, IC4 - CD4026 decade counter and 7-segment driver
- DIS1, DIS2 - LT543, 7-segment display (common-cathode)
- LED1 - 5mm IR LED
- RX1 - TSOP1738 IR receiver
- D1-D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 22-kilo-ohm
- R2, R3 - 100-ohm
- R4, R5 - 10-kilo-ohm
- VR1 - 10-kilo-ohm potentiometer

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2, C4 - 0.1µF ceramic
- C3 - 0.001µF ceramic
- C5 - 47µF, 16V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 0.5A secondary transformer
- CON1 - 2-pin plug for AC mains
- CON2, CON3 - 2-pin connector terminal
- S1 - Tactile switch

99 visitors automatically. It is built around CD4026 (decade counter and 7-segment driver in the same IC package).

Circuit and working

The circuit diagram of the infrared (IR) transmitter unit for the visitors

counter is shown in Fig. 1. It comprises transformer X1, 1N4007 rectifier diodes D1 and D2, 1000µF (25V) filter capacitor C1, 7805 (5V) voltage regulator IC1, NE555 timer IC2, IR LED (LED1) and some other components. The transmitter generates 38kHz carrier frequency.

The receiver unit with 7-segment displays is shown in Fig. 2. It comprises two CD4026 ICs (IC3 and IC4), reset switch S1, two 7-segment common-cathode displays DIS1 and DIS2, TSOP1738 IR receiver RX1 and a few other components.

Power supply for the circuit(s) is made up of transformer X1 that steps down 230V AC mains supply, full-wave rectifiers comprising diodes D1 and D2, filtering capacitor C1 and 7805 regulator IC1 to maintain a constant 5V DC at the output. The 5V DC is used by the transmitter and also fed to the receiver section by connecting CON2 to CON3.

Decade counter IC CD4026 or CD4033 (whichever is available) can be used as counter-cum-display driver.

As this project is designed for a 2-digit counter, it counts up to 99 only. The counter uses two 7-segment common-cathode displays (LT543). For one's position it uses a CD4026 (IC3) and a 7-segment common-cathode display LT543 (DIS1). For ten's position it uses another CD4026 (IC4) and another similar 7-segment display (DIS2).

The IR transmitter circuit is designed for commercial application. Its NE555 timer can be wired as a one-shot or astable multivibrator. In this circuit it is configured in astable mode. For that, we use fixed 22k resistor R1, 10k potentiometer VR1 and 0.001µF capacitor C3 for generating exact 38kHz carrier frequency.